CHUIYANG MENG

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Homepage LinkedIn Google Scholar

RESEARCH INTERESTS

Mechanistic Interpretability, Edge Intelligence, Distributed Machine Learning, Network Management.

EDUCATION

The University of British Columbia, Vancouver, Canada *September 2021 - June 2026*
Doctor of Philosophy (Ph.D.), Department of Electrical and Computer Engineering
Supervisor: Vincent W.S. Wong
Thesis: Resource-aware Distributed Edge Intelligence

Johns Hopkins University, Homewood, Baltimore, U.S.A. *August 2018 - May 2020*
Master of Science (M.S.), Department of Electrical and Computer Engineering

Harbin Institute of Technology, Harbin, P.R.C. *September 2014 - June 2018*
Bachelor of Engineering (B.Eng.), School of Electronics and Information Engineering
Supervisor: Shuai Han, Hsiao-Hwa Chen

EXPERIENCE

Simon Fraser University, Coquitlam, Canada *September 2026 - Now*
Postdoctoral Research Fellow

Vector Institute, Toronto, Canada *January 2026 - April 2026*
Applied Machine Learning Intern, Responsible AI Team

The University of British Columbia, Vancouver, Canada *September 2021 - June 2026*
Graduate Research Assistant

SELECTED PUBLICATIONS

Conference Publications

1. **Chuiyang Meng**, Wenlu Yu, Ming Tang, and Cheng Li, "Social Circuits Behind Multi-Agent Echo Chambers", submitted to *Proceedings of International Conference on Learning Representations (ICLR)*, 2027.
2. **Chuiyang Meng**, Ming Tang, and Vincent W.S. Wong, "CORTEX: Learning to Share and Specialize in Dense Language Models", submitted to *Proceedings of International Conference on Learning Representations (ICLR)*, 2027.
3. **Chuiyang Meng**, Ming Tang, and Vincent W.S. Wong, "FLoRG: Federated Fine-tuning with Low-rank Gram Matrices and Procrustes Alignment", in *Proceedings of International Conference on Learning Representations (ICLR)*, Rio de Janeiro, Brazil, April 2026, Acceptance rate: 28%.
4. **Chuiyang Meng**, Ming Tang, and Vincent W.S. Wong, "ZorBA: Zeroth-order Federated Fine-tuning of LLMs with Heterogeneous Block Activation", accepted for publication in *Proceedings of IEEE International Conference on Computer Communications (INFOCOM)*, Tokyo, Japan, May 2026, Acceptance rate: 18.9%.
5. **Chuiyang Meng**, Ming Tang, Mehdi Setayesh, and Vincent W.S. Wong, "GNN-based Neighbor Selection and Resource Allocation for Decentralized Federated Learning", in *Proceedings of IEEE Global Communications Conference (GLOBECOM)*, Kuala Lumpur, Malaysia, Dec. 2023.

6. Xiao Deng, Xuanli Wu, **Chuiyang Meng**, Shuai Han, Xiaojie Fang"A mobile relay selection strategy in cooperative spectrum sharing framework", in *Proceedings of IEEE International Conference on Communications (ICC)*, Paris, France, May 2017.

Journal Publications

1. **Chuiyang Meng**, Ming Tang, and Vincent W.S. Wong, "ASFL: An Adaptive Model Splitting and Resource Allocation Framework for Split Federated Learning", accepted for publication in *IEEE Transactions on Mobile Computing (TMC)*, early access, 2026.
2. **Chuiyang Meng**, Ming Tang, Mehdi Setayesh, and Vincent W.S. Wong, "Tackling Resource Allocation for Decentralized Federated Learning: A GNN-based Approach", in *IEEE Transactions on Mobile Computing (TMC)*, vol. 24, no. 10, pp. 9554-9569, Oct. 2025.
3. Zijun Gong, Cheng Li, Fan Jiang, Ruoyu Su, Ramachandran Venkatesan, **Chuiyang Meng**, Shuai Han, Yan Zhang, Shudong Liu, Kun Hao, "Design, Analysis, and Field Testing of an Innovative Drone-Assisted Zero-Configuration Localization Framework for Wireless Sensor Networks", in *IEEE Transactions on Vehicular Technology (TVT)*, vol. 66, no. 11, pp. 10322-10335, Nov. 2017.
4. Cheng Li, Fan Jiang, **Chuiyang Meng**, Zijun Gong, "A New Turbo Equalizer Conditioned on Estimated Channel for MIMO MMSE Receiver", in *IEEE Communications Letters*, vol. 21, no. 4, pp. 957-960, Apr. 2017.

PROFESSIONAL SERVICES

Technical Program Committee (TPC) Symposium Chair

- International Conference on Wireless Communications and Signal Processing (WCSP), Shenzhen, China

Technical Reviewers

- NeurIPS, CVPR, ICLR
- IEEE Transactions on Wireless Communications (TWC)
- IEEE Transactions on Mobile Computing (TMC)
- IEEE Transactions on Communications (TCOM)
- IEEE Internet of Things Journal (IoT-J)
- IEEE Wireless Communications Magazine

AWARDS

- IEEE INFOCOM 2026 Student Travel Grant ($1640 USD)
- President's Academic Excellence Initiative PhD Award, The University of British Columbia, 2021 - 2026
- International Tuition Award, The University of British Columbia, 2021 - 2026

TEACHING

Teaching Assistant

- **ELEC 331 Computer Communications**
  Fall 2021, Spring 2022, Fall 2023, Spring 2024, Spring 2025
  The University of British Columbia, Department of Electrical and Computer Engineering

– **EN 520.647 Information Theory**
Fall 2019
Johns Hopkins University, Department of Electrical and Computer Engineering